BRILLIANTS' CONVENT

SR. SEC. SCHOOL

WEST ENCLAVE , PITAMPURA, DELHI – 110034

BRITISH COUNCIL
International Dimension in Schools 2020–23



CLASS – IX

HOLIDAYS HOMEWORK – 2021



"We can easily manage if we will only take, each day, the burden appointed to it.
But the load will be too heavy for us it we carry yesterday's burden over again today,
and then add the burden of the morrow before we are required to bear it."
- John Newton

Dear children,

Summer vacation is a time for fun and frolic and time to bring you some constructive activities to enhance your learning process. It is probably the best time of the year for you all. Vacation is a time when you connect with your friends, family members and relatives. School work can be fun filled too! Therefore the teachers have planned some activities and projects for you to develop hands- on approach towards learning. All tasks have been designed keeping in mind your learning in all subjects while you enjoy the break.

Here are few tips for you to follow:

- Spend quality time with your parents and show that you care. Listen stories from family members and try to narrate them.
- Plan a daily routine for yourself.
- Physical fitness is an important aspect of life. A healthy mind stays in healthy body. Do yoga/ aerobics and exercise regularly to remain fit and healthy. Eat nutritious home cooked food.
- Help your mother in doing household chores. Go out of your way to do small things for your family.
- Free play time should alternate with structural games.

Note: Holiday Homework should be well labelled and submitted in the first week after summer vacations in a Pdf format.

ENGLISH



1. Based on the visual, write descriptive paragraph on this eminent personality in 150 words.
2. Write a diary entry expressing your feelings when you got the news that the schools are going to be online now and no physical classes will be there. Write the answer by taking reference from the chapter 'The Fun They Had'.
3. Make a flow chart of the chapter 'The Lost Child'. It should be informative as well.
4. Read the following stories by Ruskin Bond and write its summary with a beautiful cover page.
    I. Crazy Times with Uncle Ken

    OR

    II. The Day Grandfather Tickled a Tiger
5. Revise all the chapters and topics done in the class so far.

हिंदी

1. "आज का युग सूचना प्रौद्योगिकी (आई टी) का युग" विषय पर एक अनुच्छेद 100 से 120 शब्दों में लिखिए।

2. पाठ- दुख का अधिकार , तुम कब जाओगे अतिथि और गिल्लू (संचयन) तीनों पाठों को पूरे मन से दो बार पढ़िए और प्रत्येक पाठ के आधार पर निम्नलिखित कार्य कीजिए।

    I) प्रत्येक पाठ के बीच में से 10 शब्द चुनिए और उनके विलोम शब्द लिखिए।

    II) प्रत्येक पाठ में से 10 शब्द चुनिए और उनके दो - दो पर्यायवाची शब्द लिखिए।

    (यह कार्य अपनी व्याकरण उत्तर - पुस्तिका में ही कीजिए।

- आपका कार्य आकर्षक होना चाहिए।
- आपका कार्य आंतरिक मूल्यांकन के अंतर्गत जांचा जाएगा।

MATHEMATICS

1. Do the questions as follows in your Maths Notebook:

1. Explain each of the following in p/q form:

2. (i) 0.675 (ii) $0.3\overline{2}$ (iii) $0.12\overline{3}$ (iv) $0.003\overline{52}$ (v) $4.\overline{32}$ (vi) 2.317317317.....

3. Find two irrational numbers and two rational numbers between 0.5 and 0.55

4. Simplify each of the following by rationalizing the denominator.

5. (i) $\frac{7+3\sqrt{5}}{7-3\sqrt{5}}$ (ii) $\frac{2\sqrt{3}-\sqrt{5}}{2\sqrt{2}+3\sqrt{3}}$ (iii) $\frac{7\sqrt{3}-5\sqrt{2}}{\sqrt{48}+\sqrt{18}}$

6. Simplify:- a) $3\sqrt{5}+-\sqrt{5}+\sqrt{180}$ (b) $\sqrt{54}+\sqrt{150}$

7. Give an example each of two irrational numbers, whose

   (i) difference is a rational number
   (ii) difference is an irrational number
   (iii) sum is a rational number
   (iv) sum is an irrational number
   (v) product is a rational number
   (vi) product is an irrational number
   (vii) quotient is a rational number
   (viii) quotient is an irrational number

8. Without actual division decide which of following rational numbers have terminating decimal representation:-

(i) $\frac{33}{375}$ (ii) $\frac{15}{28}$ (iii) $\frac{16}{45}$ (iv) $\frac{12}{35}$ (v) $\frac{80}{27}$ (vi) $\frac{123}{1250}$

9. Examine whether the following numbers are rational or irrational

10. (i) $\frac{3\sqrt{8}}{\sqrt{2}}$ (ii) $\left(\sqrt{2}+\frac{1}{\sqrt{2}}\right)^2$ (iii) $\frac{22/7}{5\Pi}$ (iv) $(3+\sqrt{2})(2-\sqrt{3})(3-\sqrt{2})$

$(2+\sqrt{3})$

SCIENCE

PHYSICS

TOPIC : MOTION

1. What is meant by the statement 'Rest and motion are relative terms'? Give example to show it.
2. Explain whether the walls of a classroom are at rest or in motion.
3. Define scalar and vector quantities.
4. Identify the following as scalar or vector quantities:- mass, velocity, speed, length, distance, displacement, temperature, force, weight, power, work and energy.
5. The school of a boy from his home is 1 km to the east. When he reaches back home, he says that he had traveled 2 km distance but his displacement is zero. Justify your answer.
6. Under what condition, the average speed is equal to the magnitude of the average velocity.
7. Can the average speed of a moving body be zero?
8. Can the average velocity of a moving body be zero? State examples.
9. A car covers a distance of 5 km in 20 mins. Find the velocity of the car in
   (a) km/min (b) m/s (c) m/min (d) km/hr.

10. A train is moving with a velocity of 45km/hr. calculate the distance traveled by it in 1 hr, 1 min, 1 second.
11. An object P is moving with a constant velocity for 5 mins. Another object Q is moving with changing velocity for 5 mins. Out of these two objects, which one has acceleration? Explain.
12. Can an object be accelerated if it is moving with constant speed? If yes, explain giving examples.
13. (i) When do you say that an object has positive acceleration?
    (ii) When do you say that an object has negative acceleration?
14. State which of the following situations are possible and give an example of each of these:-
    (a) A body moving with constant acceleration but with zero velocity.
    (b) A body moving horizontally with acceleration in vertical direction.
    (c) A body moving with a constant speed in an accelerated motion.
15. What is a reference point?
16. Name the 2 physical quantities which can be obtained from velocity-time graph.
17. An electric train is moving with a velocity of 120km/hr. how much distance will it cover in 30 sec?
18. Give differences between linear motion and circular motion.

CHEMISTRY

1. Revise chapter 1 Matter in our surroundings done in the class.
2. Complete the worksheets attached.
3. Revise the activities of chapter 1 discussed in the class.

ASSIGNMENT QUESTIONS SET – 1
CHAPTER – 1
MATTER IN OUR SURROUNDINGS

1. Fill in the blanks

a. The process of _____ causes cooling.
b. The process of cooling glass is known as _____.
c. Liquids have no fixed _____ but have fixed _____.
d. _____ exists in all three states of matter.
e. Carbon dioxide is a white solid called _____ at temperature below _____.

2. State True or False

a. Evaporation of water is a bulk phenomenon.
b. Diffusion takes place in haphazard and random way.
c. SI unit of pressure is Pascal.
d. A gas is highly incompressible fluid.
e. Solids and liquids can be identified from their characteristic melting and boiling points.

3. Zig-zag movement of the solute particle in a solution is known as

(a) Linear motion
(b) Circular motion
(c) Brownian motion
(d) Curved motion.

4. Gases can be liquified by

(a) increasing pressure
(b) decreasing temperature
(c) both (a) and (b)
(d) decreasing pressure

5. Density of a substance is defined as
   (a) ratio of mass and volume
   (b) product of mass and volume
   (c) ratio of mass and temperature
   (d) product of mass and temperature

6. Which of the following is not matter
   (a) Blood
   (b) Humidity
   (c) Electron
   (d) Moon rock

7. Which is more effective in cooling?
   (a) Ice at 0° C
   (b) Water at 0° C
   (c) Water at 100° C
   (d) Ice at 100° C

8. 0° C temperature is equal to
   (a) 0 K
   (b) 273 K
   (c) -273 K
   (d) 300 K

9. The process involving the change of state from solid to gas is called
   (a) melting
   (b) boiling
   (c) sublimation
   (c) fusion

10. A solid has
   (a) definite volume and no definite shape
   (b) no definite volume no definite shape
   (c) definite shape and volume
   (d) definite shape but no definite volume

11. A liquid has
   (a) definite volume and no definite shape
   (b) no definite volume no definite shape
   (c) definite shape and volume
   (d) definite shape but no definite volume

12. A gas has
   (a) definite volume and no definite shape
   (b) no definite volume no definite shape
   (c) definite shape and volume
   (d) definite shape but no definite volume

13. Which of the following is NOT a property of particles of a matter?
   (a) The particles of matter are extremely small
   (b) The particles of matter have spaces between them.
   (c) The particles of matter are in stationery state.
   (d) The particles of matter attract each other.

14. Which of the following has minimum spaces among the particles?
   (a) Solids
   (b) Liquids
   (c) Gases
   (d) None of these

15. During summer, water kept in an earthen pot becomes cool because of the phenomenon of
   (a) diffusion
   (b) transpiration
   (c) osmosis
   (d) evaporation

16. Rate of diffusion is the fastest in
   (a) Solids
   (b) Liquids
   (c) Gases
   (d) None of these

17. Thermal conduction takes places in
(a) solids only
(b) liquids only
(c) gases only
(d) solids, liquids and gases.

18. Evaporation always causes
(a) thermal expansion
(b) Liquification
(c) Cooling down
(d) all of these

19. A change of state directly from solid to gas without changing into liquid state (or vice versa) is called
(a) Evalopartion
(b) Sublimation
(c) Diffusion
(d) Condensation

20. The rate of evaporation decreases with
(a) increase in humidity
(b) increase of temperature
(c) increase in wind speed
(d) increase of surface area

21. Expand CNG and LPG.

22. Arrange the following substances in increasing order of force of attraction between the particles. (i) milk (ii) salt (iii) oxygen

23. Why is sponge a solid though compressible?

24. Write one important characteristic of matter.

25. Why does a desert cooler cool better in a hot dry day?

26. Convert: (a) 25°C into kelvin scale (b) 500 K into celsius scale

27. Why does the smell of hot sizzling food reach you several metres away but to get the smell from cold food you have to go close?

28. What is the term used for change of solid state to liquid state?

29. Name the temperature at which solid and liquid states of matter can coexist.

30. Define evaporation.

31. "The wool being knitted into a sweater is a physical change." Justify the statement.

32. Mention two ways to liquefy atmospheric gases.

33. What is the value of boiling point of water on Kelvin Scale of temperature?

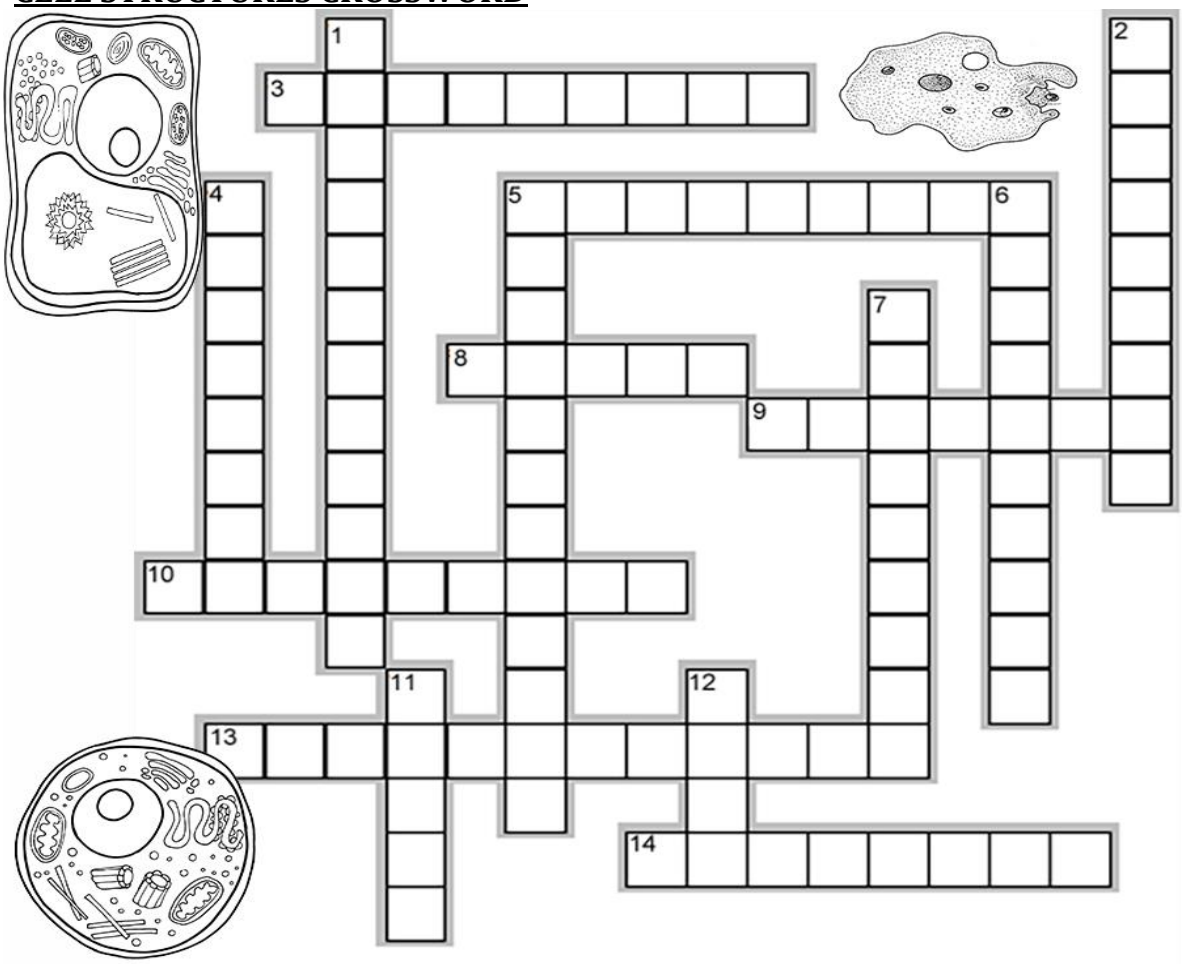
34. What is dry ice?

35. (a) Dry ice is compressed at high pressure. What happens when pressure is released?
(b) Suggest a method to liquefy atmospheric gases.

BIOLOGY

CHAPTER INVOLVED:
THE FUNDAMENTAL UNIT OF LIFE

CELL STRUCTURES CROSSWORD



Across

3. makes protein
5. interior of the cell, mostly water
8. 1st person to see cells
9. stores water; large in plants
10. transport within the cell; endoplasmic____
13. photosynthetic organelle in plants
14. tail-like structure that function sin movement

Down

1. "powerhouse"of the cell
2. organelles that break down substances
4. outer boundary of the cell
5. maintains the cell's shape
6. tool used to view cells
7. makes ribosomes
11. packaging andexporting; ____body
12. in plants, structure outside the membrane; cell___

WORKSHEET 2 (BIOLOGY)
Cells

cells	chlorophyll	microscope	sunlight
cell membrane	chloroplasts	mitochondria	vacuoles
cell wall	cytoplasm	nucleus	

Fill in the blanks.

Plants and animals are living things. All plants and animals are made of _____. The _____ is a layer around the outside of all cells. Plant cells have an additional layer called the _____ that provides extra support.

_____ is a gel-like substance inside the cell membrane.

All activity in the cell is controlled by the _____.

_____ break down food and turn it into energy for the cell to use. Water, food, and wastes are stored in the _____ of the cell. A green chemical called _____ is in the _____ of plant cells. Plant cells turn energy from _____ into food.

People didn't know that cells existed until they could see them under a _____.

S.SCIENCE

1. Revise all the chapters done in the class so far.
2. Locate and label the following places on the map of India/ World and paste all the maps in a notebook

LIST OF MAP ITEMS CLASS IX (2021-22)

SUBJECT - HISTORY

Chapter-1: The French Revolution

Outline Political Map of France (For locating and labeling / Identification)

- Bordeaux
- Nantes
- Paris
- Marseilles

Chapter-2: Socialism in Europe and the Russian Revolution

Outline Political Map of World (For locating and labeling / Identification)

- Major countries of First World War (Central Powers and Allied Powers)
- Central Powers - Germany, Austria-Hungary, Turkey (Ottoman Empire)
- Allied Powers - France, England, Russia, U.S.A.

Chapter-3: Nazism and Rise of Hitler

Outline Political Map of World (For locating and labeling / Identification)

Major countries of Second World War

- Axis Powers – Germany, Italy, Japan
- Allied Powers – UK, France, Former USSR, USA

Territories under German expansion (Nazi Power)

Austria, Poland, Czechoslovakia (only Slovakia shown in the map), Denmark, Lithuania, France, Belgium

SUBJECT – GEOGRAPHY (Outline Political Map of India)

Chapter -1: India-Size and Location

- India-States with Capitals, Tropic of Cancer, Standard Meridian (Location and Labelling)

Chapter -2: Physical Features of India

- Mountain Ranges: The Karakoram, The Zasker, The Shivalik, The Aravali, The Vindhya, The Satpura, Western & Eastern Ghats
- Mountain Peaks – K2, Kanchan Junga, Anai Mudi
- Plateau - Deccan Plateau, Chotta Nagpur Plateau, Malwa Plateau
- Coastal Plains - Konkan, Malabar, Coromandal & Northern Circar (Location and Labelling)

Chapter -3: Drainage

Rivers: (Identification only)

- The Himalayan River Systems-The Indus, The Ganges, and The Satluj
- The Peninsular rivers-The Narmada, The Tapi, The Kaveri, The Krishna, The Godavari, The Mahanadi

Lakes: Wular, Pulicat, Sambhar, Chilika

Chapter - 4: Climate

- Areas receiving rainfall less than 20 cm and over 400 cm (Identification only)13

Chapter - 5: Natural Vegetation and Wild Life

- Vegetation Type: Tropical Evergreen Forest, Tropical Deciduous Forest, Thorn

- Forest, Montane Forests and Mangrove- For identification only
- National Parks: Corbett, Kaziranga, Ranthambor, Shivpuri, Kanha, Simlipal & Manas
- Bird Sanctuaries: Bharatpur and Ranganthitto
- Wild Life Sanctuaries: Sariska, Mudumalai, Rajaji, Dachigam (Location and Labelling)

Chapter - 6: Population (Location and Labelling)

- The state having highest and lowest density of population
- The state having highest and lowest sex ratio
- Largest and smallest state according to area…

3. Prepare a project file on the topic – Disaster Management (minimum – 15 pages – maximum 20)
   - Name, Class, Name of Project, Roll no.
   - Index
   - Acknowledgement
   - Content
   - Bibliography
   - Pictures

COMPUTER

1. Make a PPT on "Artificial Intelligence Gadgets" (Minimum 10 slides)

   Points :
   - Explain Artificial Intelligence
   - Enlist any 5 gadgets based on Artificial Intelligence and their description.
2. Write python program for : (Do it in computer notebook)
   a) To print your name, class and section.
   b) To find the square root of a number.
   c) To convert kilometres into miles.
   d) To generate a random number.
   e) To swap two variables.

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